



The University of Danang

University of Science and Technology

MATHEMATICS FOR COMPUTER SCIENCE

Chap 2. . Linear Algebra



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References

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2. E. Lehman, F. T. Leighton, A. R. Meyer, 2017, **Mathematics for Computer Science**, Eric Lehman Google Inc, 998 pages
3. A. Laaksonen, **Competitive Programmer's Handbook**, 2018, 286 pages.
4. W. H. Press, S. A. Teukolsky, W.T. Vetterling, B. P. Flannery **Numerical Recipes: The Art of Scientific Computing**, Third Edition, Cambridge University Press, 1262 pages.
5. Other online/offline learning resources

Linear Algebra

- Introduction
- Eigenvectors and finding Eigenvectors
- **Matrix Decompositions**
 - Cholesky Decomposition
 - Eigendecomposition and Diagonalization
 - Singular Value Decomposition

Matrix Decompositions

- There are many ways to factorize special types of matrices that we encounter often in machine learning.
 - In the positive real numbers, we have the square-root operation that gives us a decomposition of the number into identical components, e.g., $9 = 3 \times 3$.
 - For matrices, we need to be careful that we compute a square-root-like operation on positive quantities.
 - For symmetric, positive definite matrices we can Cholesky choose from a number of square-root equivalent operations. The *Cholesky decomposition/Cholesky factorization* provides a square-root equivalent operation on symmetric, positive definite matrices that is useful in practice.

Matrix Decompositions-Cholesky Decomposition

- A symmetric, positive definite matrix A can be factorized into a product $A = LL^T$, where L is a lower-triangular matrix with positive diagonal elements:

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n1} & \cdots & l_{nn} \end{bmatrix} \begin{bmatrix} l_{11} & \cdots & l_{n1} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & l_{nn} \end{bmatrix}$$

L is called the Cholesky factor of A , and L is unique.

$$\begin{bmatrix} A_{00} & A_{01} & A_{02} \\ A_{10} & A_{11} & A_{12} \\ A_{20} & A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} L_{00} & 0 & 0 \\ L_{10} & L_{11} & 0 \\ L_{20} & L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} L_{00} & L_{10} & L_{20} \\ 0 & L_{11} & L_{21} \\ 0 & 0 & L_{22} \end{bmatrix}$$

Lower Triangular L

Transpose of L

A $n \times n$ symmetric real matrix A is said to be **positive-definite** if $x^T A x > 0$ for all non-zero $x \in \mathbb{R}^n$

Matrix Decompositions-Cholesky Decomposition

- Consider a symmetric, positive definite matrix $A \in \mathbb{R}^{3 \times 3}$. We are interested in finding its Cholesky factorization $A = LL^T$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = LL^T = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

- Multiplying out the right-hand side yields

$$A = \begin{bmatrix} l_{11}^2 & l_{21}l_{11} & l_{31}l_{11} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{31}l_{21} + l_{32}l_{22} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

Matrix Decompositions-Cholesky Decomposition

- Comparing the left-hand side and the right-hand side shows that there is a simple pattern in the diagonal elements l_{ii} :

$$A = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{21}l_{11} & l_{31}l_{11} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{31}l_{21} + l_{32}l_{22} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

$$l_{11} = \sqrt{a_{11}}, \quad l_{22} = \sqrt{a_{22} - l_{21}^2}, \quad l_{33} = \sqrt{a_{33} - (l_{31}^2 + l_{32}^2)}$$

$$l_{21} = \frac{1}{l_{11}}a_{21}, \quad l_{31} = \frac{1}{l_{11}}a_{31}, \quad l_{32} = \frac{1}{l_{22}}(a_{32} - l_{31}l_{21})$$

Matrix Decompositions-Cholesky Decomposition

- Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}.$$

$$l_{11} = \sqrt{a_{11}} = 1$$

$$l_{21} = \frac{1}{l_{11}} (a_{21} - 0) = 2$$

$$l_{22} = \sqrt{a_{22} - l_{21}^2} = \sqrt{3 - 4} \notin \mathbb{R}$$

→ A cannot be positive definite!

Find the decomposition of

$$A = \begin{pmatrix} 4 & -2 & 2 \\ -2 & 10 & -7 \\ 2 & -7 & 6 \end{pmatrix}.$$

$$l_{11} = \sqrt{a_{11}} = 2$$

$$l_{21} = \frac{1}{l_{11}} a_{21} = -1$$

$$l_{31} = \frac{1}{l_{11}} a_{31} = 1$$

$$l_{22} = \sqrt{a_{22} - l_{21}^2} = \sqrt{10 - 1} = 3$$

$$l_{32} = \frac{1}{l_{22}} (a_{32} - l_{31}l_{21}) = \frac{1}{3}[-7 - 1 \times (-1)] = -2$$

$$l_{33} = \sqrt{a_{33} - l_{31}^2 - l_{32}^2} = \sqrt{6 - 1 - 4} = 1$$

$$L = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 3 & 0 \\ 1 & -2 & 1 \end{pmatrix}$$

Matrix Decompositions-Cholesky Decomposition

- Example:

$$\begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 & 0 \\ 3 & 3 & 0 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 5 & 3 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

$$l_{11} = \sqrt{a_{11}}, \quad l_{22} = \sqrt{a_{22} - l_{21}^2}, \quad l_{33} = \sqrt{a_{33} - (l_{31}^2 + l_{32}^2)}$$

$$l_{21} = \frac{1}{l_{11}}a_{21}, \quad l_{31} = \frac{1}{l_{11}}a_{31}, \quad l_{32} = \frac{1}{l_{22}}(a_{32} - l_{31}l_{21})$$

Matrix Decompositions-Cholesky Decomposition

- A closely related variant of the classical Cholesky decomposition is the $A = LDL^T$ decomposition, where L is a lower unit triangular (unitriangular) matrix, and D is a diagonal matrix
 - If the coefficient matrix A is symmetrical but not necessarily positive definite, then the above Cholesky algorithms will not be valid. In this case, the following LDL^T factorized algorithms can be employed
 - the diagonal elements of L are required to be 1 at the cost of introducing an additional diagonal matrix D in the decomposition

Matrix Decompositions-Cholesky Decomposition

- A closely related variant of the classical Cholesky decomposition is the $A = LDL^T$ decomposition, where L is a lower unit triangular (unitriangular) matrix, and D is a diagonal matrix
 - The main advantage is that the LDL decomposition can be computed and used with essentially the same algorithms, but avoids extracting square roots
 - The “normal” Cholesky decomposition is obtained from this via

$$LDL^T = (LD^{\frac{1}{2}})(LD^{\frac{1}{2}})^T.$$

[The square root of $D = \text{diag}(d_1, d_2, \dots, d_n)$ is $D^{\frac{1}{2}} = \text{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})$.]

Matrix Decompositions-Cholesky Decomposition

- Given the classical Cholesky decomposition $A = LL^T$ of a positive definite matrix, if S is a diagonal matrix that contains the main diagonal of L then A can be decomposed as $A = KDK^T$ where
 - $K = LS^{-1}$ (this rescales each column to makes diagonal elements 1)
 - $D = S^2$
 - If A is positive definite then the diagonal elements of D are all positive. For positive semidefinite A , an LDL^T decomposition exists where the number of non-zero elements on the diagonal D is exactly the rank of A . Some indefinite (negative semidefinite) matrices for which no Cholesky decomposition exists have an LDL^T decomposition with negative entries in D

Matrix Decompositions-Cholesky Decomposition

- A closely related variant of the classical Cholesky decomposition is the $A = LDL^T$

$$\begin{aligned}
 A = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix} \begin{bmatrix} 1 & l_{21} & l_{31} \\ 0 & 1 & l_{32} \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} d_1 & d_1 l_{21} & d_1 l_{31} \\ d_1 l_{21} & d_2 + d_1 l_{21}^2 & d_2 l_{32} + d_1 l_{21} l_{31} \\ d_1 l_{31} & d_1 l_{21} l_{31} + d_2 l_{32} & d_1 l_{31}^2 + d_2 l_{32}^2 + d_3 \end{bmatrix}
 \end{aligned}$$

Matrix Decompositions-Cholesky Decomposition

- Example

$$\begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 6 & 1 & 0 \\ -8 & 5 & 3 \end{pmatrix} \begin{pmatrix} 2 & 6 & -8 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -4 & 5 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} 1 & 3 & -4 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad S^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \quad LS^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 1 & 0 \\ -8 & 5 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -4 & 5 & 1 \end{bmatrix}$$

- $K = LS^{-1}$

- $D = S^2$

Matrix Decompositions-Cholesky Decomposition

- The Cholesky decomposition is an important tool for the numerical computations underlying machine learning.
- Symmetric positive definite matrices require frequent manipulation, e.g., the covariance matrix of a multivariate Gaussian variable is symmetric, positive definite.

Matrix Decompositions-Cholesky Decomposition

- The Cholesky factorization of this covariance matrix allows us to generate samples from a Gaussian distribution. It also allows us to perform a linear transformation of random variables, which is heavily exploited when computing gradients in deep stochastic models, such as the variational auto-encoder
- The Cholesky decomposition also allows us to compute determinants very efficiently.
 - Given the Cholesky decomposition $A = LL^T$, we know that $\det(A) = \det(L) \det(L^T) = \det(L)^2$.
 - Since L is a triangular matrix, the determinant is simply the product of its diagonal entries so that $\det(A) = \prod_i l_{ii}^2$. Thus, many numerical software packages use the Cholesky decomposition to make computations more efficient.

Linear Algebra

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